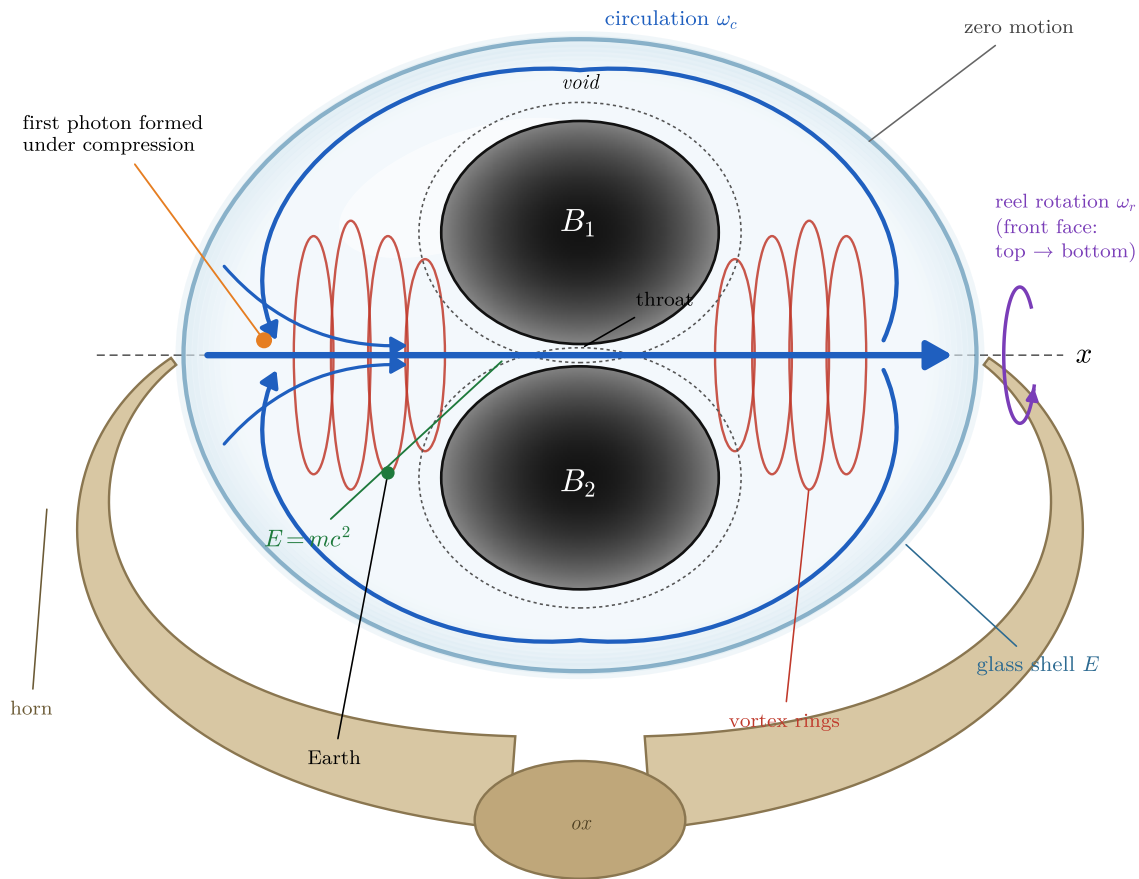
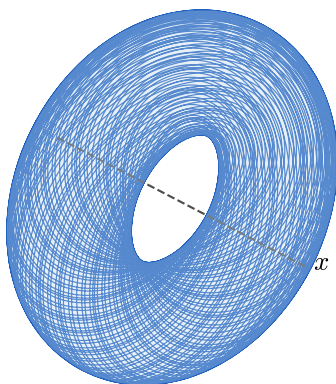


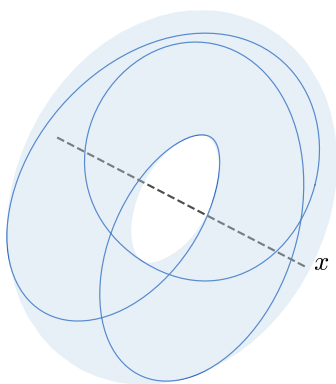
(a)



(b)



(c)



$\omega_r/\omega_c = (1 + \sqrt{5})/2$ (irrational): dense, never closes

$\omega_r/\omega_c = 3/2$ (rational): closes after two circuits